

Introducing ApparelFlow ERP: Transforming Apparel Manufacturing Management

We are thrilled to unveil **ApparelFlow ERP** — a modern, full-stack ERP system engineered specifically to streamline garment production workflows, enforce strict quality controls, and bring macro-level visibility to factory managers and floor supervisors alike.

Core Highlights & Architectural Features

Admin Style Recipe Management Workbench

Located directly within the System Admin Dashboard, the Recipe Management workbench provides full lifecycle control over apparel production specifications:

- **Recipe Registry View:** Displays active style recipes (e.g., Casual Blouse, Crop Top) with fabric types, target yardage, component counts, and wastage caps.
- **Interactive Recipe Creator & Editor:** Configure style codes, fabric types, standard yards per garment, maximum wastage thresholds, and build dynamic garment component piece specifications.
- **Delete Protection Guard:** Prevents orphaned production runs by blocking deletions of style recipes tied to active cutting batches.
- **Atomic Backend APIs:** RESTful endpoints supporting nested component transactions and server-enforced data consistency.

4 Specialized Departmental Dashboards

ApparelFlow ERP separates operational workflows across role-tailored workstations without component fragmentation:

1. Cutting Supervisor Dashboard (/cutting-supervisor): Initiate cutting batches with recipe yardage calculations, track active cutting progress, review QA feedback notes, and dispatch orders directly to verification.

2. Cutting Verifier Dashboard (/verification): Features an interactive audit terminal with real-time traffic-light indicators (● Green, ● Yellow, ● Red) and a server-enforced hard stop that prevents order approval whenever piece count shortages are detected.

3. Sewing Supervisor Dashboard (/sewing): Isolated assembly queue displaying only verified orders, complete with immutable QC audit cards, color-coded fabric wastage analytics, and line-assignment controls.

4. System Admin Supreme Dashboard (/admin): Macro control center with plant-wide KPI banners, real-time cross-departmental order tracking, omni-access departmental links, and full staff roster administration.

Middleware RBAC & Seamless Navigation

- **Middleware RBAC Guard:** Ensures users are restricted strictly to authorized departmental routes while granting admins full omni-access.
- **Root Router & 1-Click Role Switcher:** Automatically routes users to their workstation upon login and includes an integrated role switcher for rapid evaluation across all 4 roles.

Build Status & Deployment Readiness

The application compiles cleanly with zero TypeScript or routing errors across all 13 production routes:

None

Route (app)

```
├─ ○ /
├─ ○ /admin
├─ f /api/orders
├─ f /api/orders/[id]
├─ f /api/orders/[id]/verify
```

```
└─ f /api/recipes
└─ f /api/sewing/queue
└─ f /api/stats
└─ o /cutting-supervisor
└─ o /dashboard
└─ o /login
└─ o /sewing
└─ o /verification
✓ Generating static pages (13/13) compiled successfully
```